

Assignment 3

Responsive Web Design · Media Queries + Bootstrap Grid

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Source repository: <https://github.com/mmionya/web-technologies-assignment3>

Project and setup

This project is a simple Jamie Paige fan page. It has four sections: an introduction, About the music, albums and frequently asked questions. The page uses local album images, a plain font and standard Bootstrap components.

index.html connects Bootstrap 5.3.8 CSS from its CDN, followed by style.css. The viewport meta tag sets the layout width to the device width. The Bootstrap JavaScript bundle appears before the closing body tag and powers the menu and accordion.

Task 1 · A section with custom media queries

The About the music section (#values) uses only custom classes. Its .values-grid starts with one column. At min-width: 768px it changes to two columns; at min-width: 1200px it changes to three. At 768px, the heading grows from 1.5rem to 1.75rem and the gap grows from 1rem to 1.5rem. This is mobile-first because the phone layout works before either media query applies.

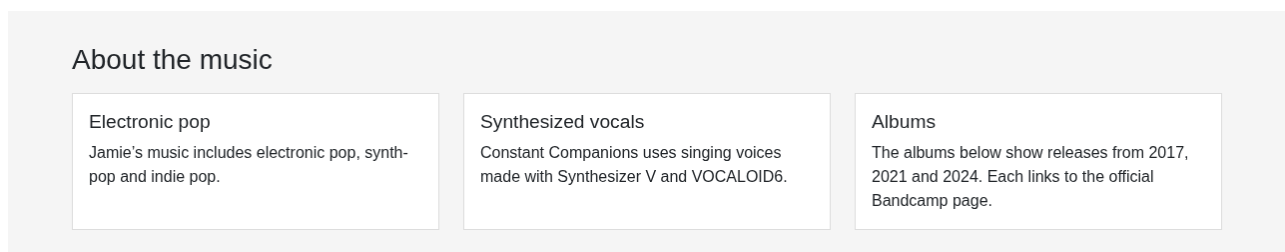


Figure 1. Three custom CSS columns at a 1280 px viewport.

Technical references: <https://getbootstrap.com/docs/5.3/getting-started/introduction/> and <https://getbootstrap.com/docs/5.3/layout/grid/>. Artwork credits and official album sources are recorded in assets/CREDITS.md; README.md includes the component documentation links.

Task 1 · Phone and tablet evidence

The three panels describe electronic pop, synthesized vocals and albums. At 375 px, each panel has its own row. At 768 px, two panels share the first row and the third continues below. CSS Grid shares the available space without fixed panel widths or absolute positioning.

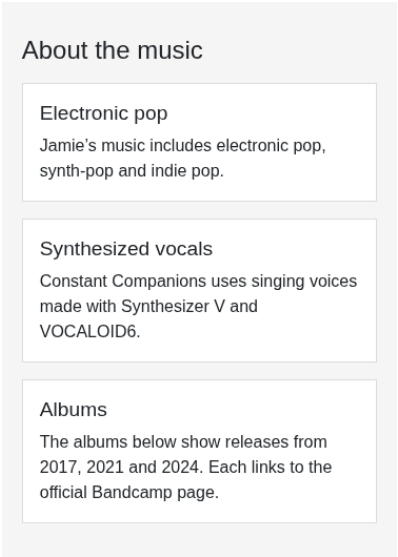


Figure 2. 375 px: one column.

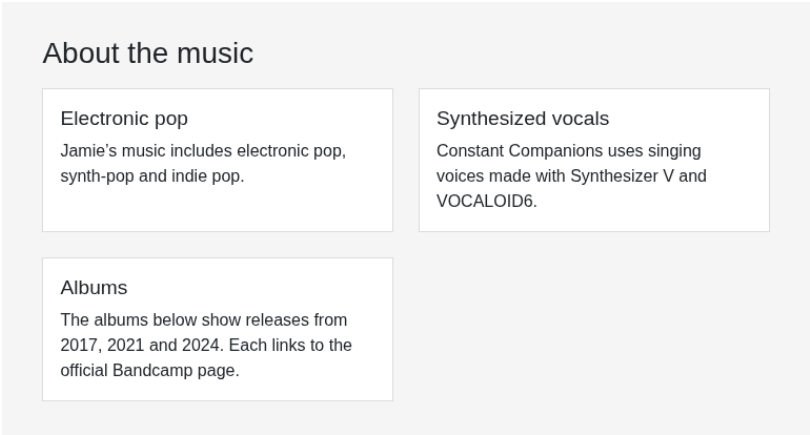


Figure 3. 768 px: two columns.

Task 2 · Bootstrap grid

The album section uses .container, .row and responsive columns: .col-12 .col-md-6 .col-xl-4. Bootstrap's 12-column grid therefore shows one album at 375 px, two at 768 px and three at 1280 px. Bootstrap cards keep the artwork, titles and links together; spacing utilities add consistent gaps.

Albums

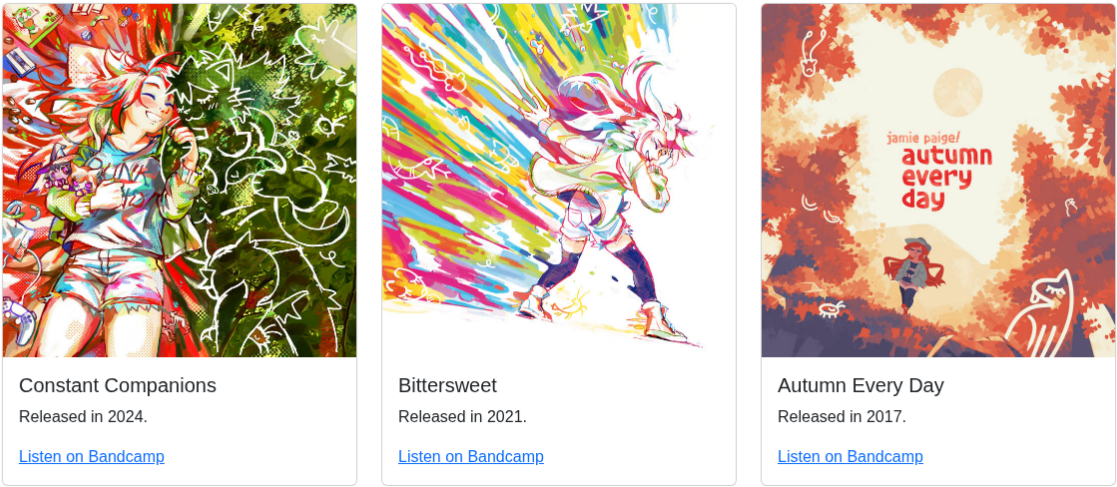


Figure 4. Three Bootstrap album cards at 1280 px.

The introduction uses two Bootstrap columns on wider screens. Responsive order-* classes put the introduction text before the artwork on a phone, so the purpose of the page appears first. On a desktop, the columns sit beside each other.

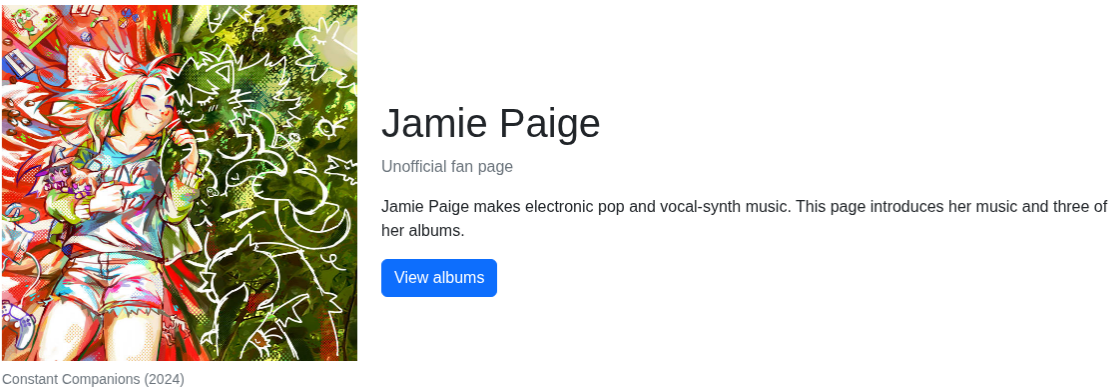


Figure 5. The introduction's two-column desktop layout.

Task 2 · Mobile order and interactive components

The .navbar-expand-lg navigation collapses below 992 px. Its menu button shows links to the page sections. The Unofficial fan page note uses .d-none .d-md-block, so it is hidden below 768 px. The FAQ is a Bootstrap accordion: its buttons show answers using data-bs-* attributes. The page does not need custom JavaScript.

These phone captures show the introduction with text first, the expanded menu, and an open answer. Browser checks also verify all three required viewport widths, both grids, section links, interactive controls and the absence of horizontal scrolling.

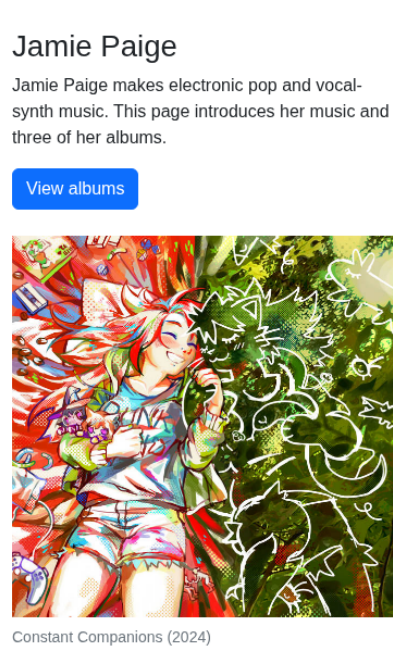


Figure 6. Text comes first at 375 px.

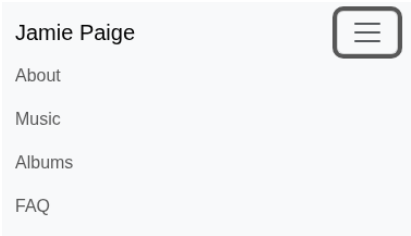


Figure 7. The expanded mobile navigation.

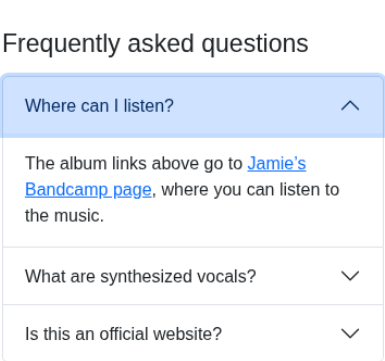
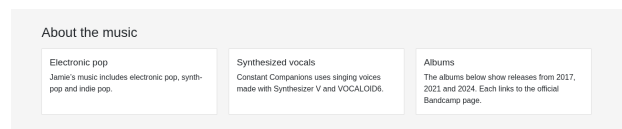
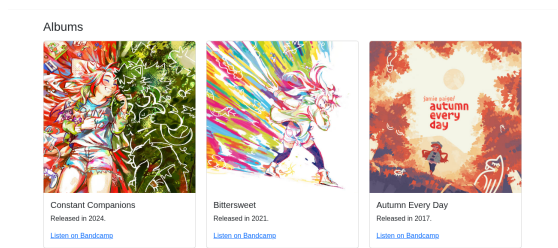


Figure 8. An expanded FAQ answer.

Task 3 · Comparison of the two approaches



Custom CSS at 1280 px (detail: Figure 1).



Bootstrap at 1280 px (detail: Figure 4).

Aspect	Custom CSS: About the music	Bootstrap: albums and introduction
Responsive code	The .values-grid rule defines the base layout; two min-width media queries change its column count.	The album columns use col-12 col-md-6 col-xl-4. Bootstrap already defines the breakpoint rules.
Design control	The heading size and gap change in my own media query in style.css.	Bootstrap supplies column widths, gutters and the standard card appearance. I use its spacing classes.
Changing the layout	To change the desktop count, edit grid-template-columns in the 1200 px media query.	To change the desktop count, edit each album's col-xl-* class; the CSS framework supplies the widths.
Built-in behavior	CSS Grid arranges the three information panels. It does not provide an interactive menu or FAQ.	The Bootstrap bundle handles navbar collapse and accordion state through data attributes.

I would use custom media queries for a small section like About the music, where I want to set the columns and gaps directly. I would use Bootstrap for a whole page that needs common layouts and components, such as the album cards, menu and FAQ. Using both here lets me practise writing responsive CSS and using a framework.

Conclusion

This page shows how a mobile-first layout grows from one column to several columns as more space becomes available. The main difference is where the responsive rules live: in my own media queries or in Bootstrap's predefined classes. Testing at 375, 768 and 1280 px also makes it easier to catch ordering problems, overflow and controls that are difficult to use on a phone.

Validation: tests/check_page.py passed at 320, 375, 768, 1024, 1280 and 1440 px, including both grids, section links, menu and accordion keyboard controls, image loading and no horizontal overflow.

Whole-page screenshot · 375 px

Captured from the implemented page in a real browser. The entire page is included; use the task figures for close-up details.

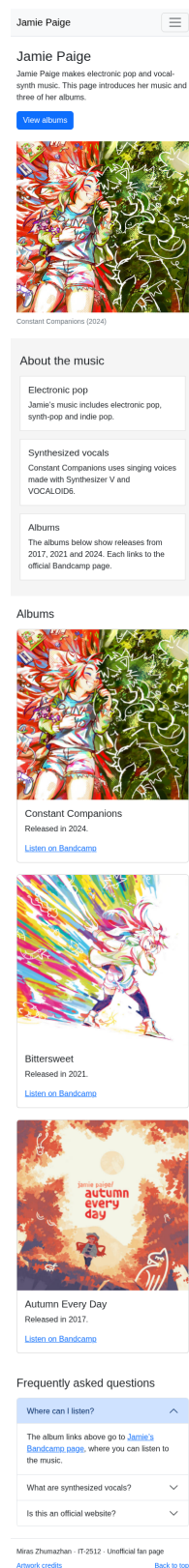


Figure 9. Complete page at a 375 px viewport.

Whole-page screenshot · 768 px

Captured from the implemented page in a real browser. The entire page is included; use the task figures for close-up details.

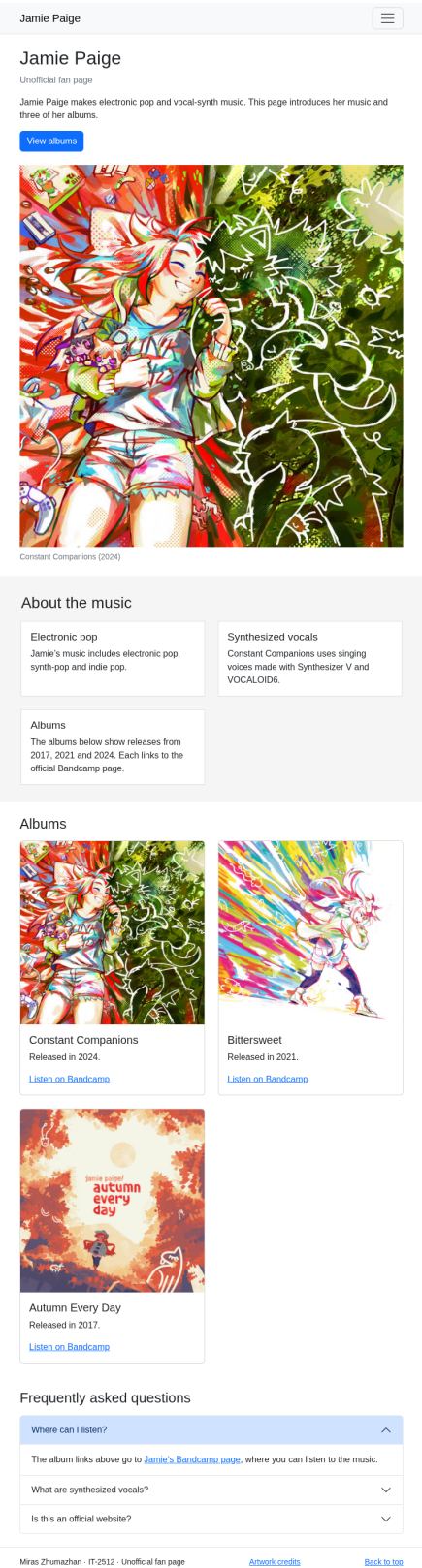


Figure 10. Complete page at a 768 px viewport.

Whole-page screenshot · 1280 px

Captured from the implemented page in a real browser. The entire page is included; use the task figures for close-up details.

Jamie Paige

AboutMusicAlbumsFAQ



Constant Companions (2024)

Jamie Paige

Unofficial fan page

Jamie Paige makes electronic pop and vocal-synth music. This page introduces her music and three of her albums.

View albums

About the music

Electronic pop

Jamie's music includes electronic pop, synth-pop and indie pop.

Synthesized vocals

Constant Companions uses singing voices made with Synthesizer V and VOCALOID6.

Albums

The albums below show releases from 2017, 2021 and 2024. Each links to the official Bandcamp page.

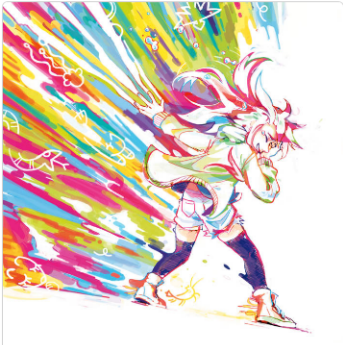
Albums



Constant Companions

Released in 2024.

Listen on Bandcamp



Bittersweet

Released in 2021.

Listen on Bandcamp



Autumn Every Day

Released in 2017.

Listen on Bandcamp

Frequently asked questions

Where can I listen?

The album links above go to [Jamie's Bandcamp page](#), where you can listen to the music.

What are synthesized vocals?

Is this an official website?

Miras Zhumazhan · IT-2512 · Unofficial fan page

Artwork credits

Back to top

Figure 11. Complete page at a 1280 px viewport.